Akshat Sanghvi

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PROFESSIONAL EXPERIENCE

Tesla | Data Scientist — Nov 2023 – Present

- Implemented an ensemble model **(ARIMA, Prophet, Croston)** to forecast demand for 26,000 SKUs across 11 distribution centers, improving predictive accuracy by 18% over previous statistical methods. Automated model deployment and updates via **CI/CD pipelines**
- Optimized code runtime by **92% (18 hours to 1.5 hours)** through parallel processing in Python, improving efficiency and scalability
- Developed an AI chatbot using **Llama API** and **RAG** pipelines, optimizing text processing with **FAISS** for 60% faster retrieval, and deploying real-time Text-to-SQL capabilities for **100+ concurrent users**
- Led the development of an automation pipeline using **Python** to track data patterns, mentoring an intern and reducing manual workload by 50% through automated **anomaly detection** and notifying stakeholders of discrepancies via email alerts
- Created **5+ Streamlit tools** to monitor key financial and operational metrics, supporting 8 teams globally and providing product leaders and executives with actionable insights for informed decision-making

Gies Business School | Graduate Research and Teaching Assistant — Jan 2022 – Nov 2023

- Conducted research with Prof. Aravinda and an Indian NGO to promote rural girls' education. Analyzed pre- and post-intervention surveys using t-tests, revealing a **36% increase** in parental agreement to support education, demonstrating the project's success
- Utilized Spark to analyze 1M+ credit consumers' data with 200+ attributes. Deployed a credit default classification model on **AWS (SageMaker)** using Random Forest and Decision Tree models in **PySpark MLlib**, achieving 60% accuracy
- Mentored 6 teams for course ECE 445 – Senior Design Lab and responsible for entire course cycle for 200+ students

Walmart | Data Science intern — May 2022 – Aug 2022

- Experimented with ARIMA and XGBoost to forecast Cases per Trailer for Outbound Supply Chain Optimization team, resulting in **$1M annual savings** and **4,320 man-hours saved per year**. Achieved 6% MAPE for non-seasonal and 12% for seasonal items
- Designed an interactive **Tableau** dashboard to track model performance KPIs ensuring effective stakeholder communication and reporting

Supercut Engineering Works | Data Scientist — Nov 2019 – Dec 2020

- Built a **customer segmentation** model using **K-Means Clustering** algorithm for targeted marketing. The initiative led to a 10% rise in the customer acquisition rate in Q4 2020 compared to Q3 2020
- Developed 3 interactive dashboards in **Tableau** & **PowerBI** to monitor production KPIs across multiple manufacturing plants, reducing reporting time by 40% and enhancing decision-making for operations teams

TECHNICAL SKILLS

- **Languages/Libraries:** Python (Pandas, NumPy Scikit-Learn, PyTorch, Tensorflow), R, C/C++, MATLAB, SAS, SQL, NoSQL
- **Framework/Technologies:** AWS, CI/CD pipelines, Git, GCP BigQuery, Docker, Kubernetes, Apache Spark, Hadoop, Tableau, PowerBI
- **Concepts:** ETL pipeline architecture, A/B Testing, Hypothesis Testing, Predictive Modeling, Machine Learning, Neural Networks

ACADEMIC PROJECTS

Image Captioning using Transformer [Link] | Python, **PyTorch**

- Applied image rotation as a pretext task to train a **ResNet18** encoder, yielding robust features for image captioning
- Built a **Transformer** decoder with custom positional encoding and attention layers, trained on 30,000 images to generate captions

Generative Adversarial Networks [Link] | Python, PyTorch

- Implemented **LSGAN** and **DCGAN** architectures from scratch to generate high-quality synthetic images
- Optimized network architectures and hyperparameters to improve GAN training stability and image quality

Land Cover and Crop Type Segmentation [Link] | Python, **TensorFlow**

- Used TensorFlow to create pixel-level labels based on crop-type maps from Cropland Layer images provided by USDA
- Incorporated **UNet** to segment Corn, Soybeans from other crops on RapidEye Satellite image with 85% Pixel accuracy

EDUCATION

University of Illinois at Urbana-Champaign — Champaign, Illinois
Master of Science in Industrial Engineering, Concentration – Advanced Analytics — May 2023

University of Mumbai — Mumbai, India
Bachelor of Engineering in Mechanical Engineering — July 2021